

Prevalence and Risk Factors for Impaired Glucose Regulation among First-degree Relatives of Patients with Type 2 Diabetes Mellitus in Maiduguri, Northeastern Nigeria

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Abstract

Background/Purpose: Owing to genetic predilection of type 2 diabetes mellitus (T2DM), the risk of developing impaired glucose metabolism is thought to be higher in first-degree relatives (FDRs) of those with T2DM. The aim of this study was to estimate the prevalence of impaired glucose tolerance (IGT) and impaired fasting glycemia (IFG) and its associated risk factors in FDRs of patients with T2DM. **Materials and Methods:** This is a cross-sectional descriptive study of 320 cases (FDRs of T2DM outpatients) aged ≥ 25 years (145 males and 175 females) and 160 controls (76 males, 84 females) who were age and sex matched. All subjects underwent anthropometric, physical activity, and laboratory assessments after an overnight fast. Oral glucose tolerance test with 75 g anhydrous glucose was administered. Fasting plasma glucose of 6.1–6.9 mmol/L and 2-h post-fasting glucose value of 7.8–11.0 mmol/L were used to define IFG and IGT, respectively. Statistical analysis was carried out using the Statistical Package for the Social Sciences software, version 20 (SPSS, Chicago, Illinois), and P value of <0.05 was considered significant. **Results:** The mean (standard deviation) age of the cases and controls was 38.4 (12.3) and 38.9 (10.3) years, respectively, $P = 0.66$. The prevalence rates of both IGT and IFG in cases and controls were (28.1% vs. 18.1%, $P = 0.019$ and 10.3% vs. 5.6%, $P = 0.0001$). The prevalence of IGT and IFG was found to be higher among females, $P < 0.05$. Multivariate analysis revealed abnormal waist circumference, being FDR, and elevated systolic blood pressure as risk factors for both the IGT and IFG, $P < 0.05$. **Conclusion:** FDRs of patients with T2DM are at higher risk of IGT or IFG. The risk increases with the development of obesity and elevated blood pressure in them.

Keywords: First-degree relatives, impaired fasting glucose, impaired glucose tolerance, risk factors, Nigeria

INTRODUCTION

Type 2 diabetes mellitus (T2DM) results from a defect in insulin secretion and impairment of insulin action in hepatic and peripheral tissues, especially muscle tissue and adipocytes.^[1] A post-receptor defect is also present, causing resistance to the stimulatory effect of insulin on glucose use.^[1] It is an important public health problem worldwide, and its prevalence is increasing in both developed and developing nations.^[2] Family members of patients with diabetes are at higher risk of developing

diabetes, although genetic factors play a key role in the development of T2DM, in the majority of patients, diabetes is brought about by a combination of genetic and environmental factors.^[3–6] With the increasing prevalence of diabetes mellitus worldwide,^[1] the number of first-degree relative (FDR) of patients with T2DM and thus an increased risk of developing diabetes will also increase,

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which means that identifying risk factors associated with susceptibility to diabetes becomes increasingly important.

Although much is known about the impact of impaired glucose regulation on FDR in developed nations, few studies have been undertaken in developing nations including Nigeria. Studies regarding the prevalence of impaired glucose tolerance (IGT), impaired fasting glycemia (IFG), and its associated risk factors in FDR of patients with diabetes are important to gain a better understanding of the disease and if possible, to prevent or delay its progression and complications in developing countries.

The objective of this study was to estimate the prevalence of IGT, IFG, and its associated risk factors in FDR of people with type 2 diabetes and compare with controls of the same sex and age.

MATERIALS AND METHODS

Subjects

A total of 320 (145 males and 175 females) relatives of 160 patients with type 2 diabetes were recruited (two relatives per one type 2 diabetes proband) as cases from a consecutive sample of patients with T2DM attending outpatient clinics at the University of Maiduguri Teaching Hospital, Maiduguri, Borno State, Nigeria. A total of 160 persons (76 males and 84 females) were recruited as controls from a group of apparently healthy volunteers (not known to be hypertensive or diabetic and with no family history of diabetes). Other exclusion criteria include the following:

- Patients previously diagnosed with diabetes
- Patients on antidiabetic treatment
- Patients on drugs such as beta-adrenergic receptor agonist, diuretics, lipid-lowering medications, and systemic steroids or other agents known to affect glucose tolerance
- Acute or chronic illness from history and physical examination
- Pregnancy, subjects who were >4 weeks amenorrheic from last menstrual period

The sample was recruited between March 2009 and March 2010. Review and authorization of all aspects of the study were carried out by the ethics committee of the institution. Approval was obtained from the committee before commencement of the study, and a declaration of informed consent was signed by each participant.

The cases (FDR of patients with T2DM) included siblings, children, or parents, probands were the persons with T2DM whose FDR is studied. All the subjects reported to the clinic in the morning after an overnight fast (8–12h), they were asked to abstain from vigorous exercise and smokers were encouraged to abstain from smoking in the evening and morning before examination.

The questionnaire to obtain personal demographic information was administered to those who cannot read or write, whereas the literate among them were able to do that by themselves. Information concerning physical activities, both in active and leisure states, of the study population was sought and scored on the pro forma for the purpose of grouping them into the following different classes of activities as defined:

Work-related physical activity was separated into three grades according to occupation as defined by the National Noncommunicable Disease Survey:

1. Not active (sedentary) (e.g., office work and unemployment)
2. Moderately active (e.g., housework, trade work, and nursing)
3. Very active (e.g., laboring)

Leisure time physical activity was graded as follows:

1. Not active (e.g., housebound)
2. Moderately active (e.g., gardening, walking, and sports 1–2 days/week)
3. Very active (e.g., sports 3 days/week)

Procedure

On arrival at the clinic, the information on family history in the questionnaire completed by the subjects was verified. Height was measured to the nearest 0.1 cm using standard graduated height scale with subjects in the erect position and without footwear or headwear.^[7,8] Body weight in kilograms (kg) was measured to the nearest 0.1 kg with subjects wearing light clothing and without shoes, using a standard hospital weighing scale positioned on a flat horizontal surface. Body mass index (BMI) (kg/m²) was calculated as weight (in kg) divided by the square of the height (in m²). Subjects with BMI < 18.5 kg/m² were classified as underweight and those with BMI between 18.5 and 24.9 kg/m² were classified as having normal weight. Those with BMI of 25.0–29.9 and ≥30.0 kg/m² were classified as overweight and obese, respectively.^[8,9]

Waist circumference (WC) was measured to the nearest 0.1 cm with tape measure placed midway between the lower rib margin and the iliac crest with the patient breathing normally.^[8] WC of ≥102 cm in males and ≥88 cm in females was considered abnormal.^[8] Hip circumference (HC) (in cm) was measured to the nearest 0.1 cm as the horizontal level of maximum circumference around the buttocks (posteriorly) and the pubic symphysis (anteriorly).^[8] Waist-to-hip ratio (WHR) was calculated as the ratio of the WC (in cm) divided by the HC (in cm). WHR of ≥ 0.90 in males and ≥0.85 in females was considered abnormal.^[8] Resting blood pressure (BP) was measured after subjects had been seated for 10 min by using a mercury sphygmomanometer and appropriately sized cuffs, using a standard technique.

Fasting venous blood samples were collected from an antecubital vein of each study and control subjects using aseptic technique, and emptied into fluoride and plain bottles for glucose and serum lipids, respectively. The samples were transferred to the laboratory immediately, promptly centrifuged, and analyzed on the same day for the estimation of fasting glucose and lipid profiles (total cholesterol, low-density lipoprotein [LDL] cholesterol, high-density lipoprotein [HDL] cholesterol, and triglyceride) using glucose oxidase and enzymatic methods. Study and control subjects were then given 75 g of anhydrous glucose dissolved in 300 mL of bottled water to drink within 5 min. Blood samples were obtained 2 h after the ingestion of oral glucose for post-glucose load plasma glucose estimation. Those with fasting plasma glucose concentration between 6.1 and 6.9 mmol/L after 8–12 h fast were considered to have IFG.^[10] Plasma glucose concentration of 7.8–11.0 mmol/L after standard oral glucose load was considered IGT.^[11] Blood glucose concentration of <7.8 mmol/L 2 h after 75 g oral glucose and/or fasting plasma glucose $\leq$ 6.0 mmol/L was considered normal glucose tolerance. All the blood sampling processes were performed in the central laboratory of the University of Maiduguri Teaching Hospital using an enzyme-linked method.

Statistical analysis

All data collected were analyzed using the Statistical Package for the Social Sciences statistical program, version 20 (SPSS, Chicago, Illinois). Mean $\pm$ standard deviation (SD) values were used to describe continuous variables and proportions for categorical data. Statistical method used included Student's *t*-test, chi-squared test, and multiple logistic regressions.

Two-tailed Student's *t*-test was used for comparison of mean values and chi-squared (χ^2) test for proportions. Fisher's exact test was used where cells contained values less than five. Comparison of proportion in ordered categories was made using the chi-squared test for linear trend.

Multivariate analysis using multiple logistic regressions was used to determine the independent risk factors for IGT and IFG. The following covariates were considered in the multivariate adjusted analyses: age, gender, being FDR, BMI, WC, HC, and systolic blood pressure (SBP) and diastolic blood pressure (DBP).

In all statistical comparisons, probability (*P*) value of <0.05 was considered significant.

RESULTS

This was a hospital-based cross-sectional study to estimate the prevalence of IGT and IFG and its associated risk factors in FDRs of people with T2DM (cases) attending the diabetes clinic of University of Maiduguri Teaching

Hospital Maiduguri, Borno State, Nigeria, compared with that of the controls from among a group of apparently healthy volunteers with no family history of diabetes or hypertension.

Of the 320 cases (145 males [45.3%] and 175 females [54.7%]), 173 had normal oral glucose tolerance test (101 males [58.4%] and 72 females [41.6%]), 24 (7.5%) had diabetes (6 males and 18 females), 90 (28.1%) had IGT (29 males and 61 females), and 33 (10.3%) had IFG (9 males and 24 females).

The prevalence of IGT increased with age in both study (χ^2 for trend = 70.13, *P* = 0.0001) and control (χ^2 for trend = 18.1, *P* = 0.0013 [Fisher's exact result]) subjects, with peak in the age group of 35–44 years in cases, whereas in controls, it peaked in the age group of 45–54 years. A decline in the prevalence of IGT was noted in the age group of 65–74 and 55–64 years in cases and controls, respectively.

The prevalence of IGT and IFG in cases and controls was 28.1% versus 10.3% and 18.1% versus 5.6%, respectively. The prevalence of IGT in cases was statistically higher than in controls (*P* = 0.019); similarly, the prevalence of IFG was statistically higher among cases than that in controls (0.0001).

The prevalence of IGT was found to be higher among females than that in males in both cases (34.9% vs. 20%, *P* = 0.004) and control (21.4% vs. 14.5%, *P* = 0.44) subjects.

Risk factors

To determine the influence of potential factors on IGT and IFG, multivariate analysis was used to determine the independent risk factors in the study population.

Age, WC, WHR, FDR, DBP, and SBP were identified independent risk factors for IGT. However, WC, WHR, and being FDR were stronger independent predictors for IGT while being FDR, SBP, and WC were the identified risk factors for IFG as shown in Table 1.

Exercise and prevalence of impaired glucose tolerance in cases and control subjects

The prevalence of IGT increased with decreasing level of physical activity in both cases (χ^2 for trend = 4.17, *P* = 0.041) and control (χ^2 for trend = 9.8, *P* = 0.002 [Fisher's exact result]) subjects.

The observed prevalence of IGT was found to be significantly higher among the inactive (18.1%) than that in the very active (0.6%) study subjects, as shown in Table 2. The difference was statistically significant (*P* = 0.005). Similarly, the prevalence of IGT was 15.6% in the inactive versus 0% in the very active controls (*P* = 0.0001), as shown in Table 2.

Similarly, the prevalence of IFG as shown in Table 2, was higher in case subjects who engaged in sedentary lifestyle (6.3%) compared with those who were very active (0.3%), $P < 0.05$. Impaired fasting glucose was also found to be

significantly higher among the inactive control subjects (4.4%) compared with the very active group ($P < 0.05$) as in Table 2.

No significant difference was observed in the mean values of total cholesterol, HDL cholesterol, LDL cholesterol, and triglycerides in case and control subjects ($P > 0.05$) as shown in Table 3.

However, the distribution of serum lipids in cases and controls according to the presence of IGT showed the mean (SD) of total cholesterol (4.9 [0.88]) and LDL cholesterol (2.9 [0.82]) to be significantly higher among case subjects with IGT compared to those without IGT ($P = 0.0001$ and 0.002 , respectively) as shown in Table 3. The mean (SD) triglyceride was also found to be higher in case subjects with IGT (1.3 [0.52]) than in those without IGT (1.2 [0.54]) though the difference was not statistically significant ($P = 0.13$). In case subjects with IGT, the mean (SD) HDL cholesterol was found to be lower (1.4 [0.42]) than in those without (1.5 [0.45]) IGT, the difference was, however, not significant ($P = 0.070$) as shown in Table 3.

In control subjects with IGT, the mean (SD) values of total cholesterol ($P = 0.041$), LDL cholesterol ($P = 0.32$), and triglyceride ($P = 0.004$) were higher than in controls without IGT. The mean (SD) HDL cholesterol, however, was found to be the same in controls with and without IGT ($P = 1.000$) as shown in Table 3.

DISCUSSION

In this cross-sectional study, cases (FDR of T2DM) showed increased prevalence of IGT, IFG, and eventually developing diabetes. In most cases, this is accompanied by unfavorable BMI, WC, lifestyle, BP, and lipid profile. With

Table 1: Odds ratio estimates for impaired glucose tolerance/ impaired fasting glycemia from multiple logistic regression

Risk factors	Odds ratio (OR)	95% CI	P value
WC			
IGT	10.97	5.4–22.3	0.0001
IFG	9.50	2.84–31.9	0.00013
WHR			
IGT	7.80	1.1–58.0	0.0005
IFG	3.80	2.7–18.9	0.0007
FDR			
IGT	3.70	1.08–29.1	0.0016
IFG	4.70	7.5–16.9	0.015
Age			
IGT	1.77	1.10–2.91	0.017
IFG	1.73	0.86–4.5	0.087
SBP			
IGT	2.99	1.07–12.7	0.019
IFG	2.94	0.92–20.7	0.035
DBP			
IGT	2.35	1.05–7.34	0.024
IFG	2.90	0.62–13.5	0.22
BMI			
IGT	1.70	0.85–4.0	0.093
IFG	1.58	0.64–5.6	0.21

CI= confidence interval, IGT = impaired glucose tolerance, IFG = impaired fasting glycemia, WC = waist circumference, WHR = waist-to-hip ratio, FDR = first-degree relative, SBP = systolic blood pressure, DBP = diastolic blood pressure, BMI = body mass index

Table 2: Level of physical activity and prevalence of impaired glucose tolerance/impaired fasting glycemia in the case/control populations

Level of physical activity	IGT		IFG	
	Case	Control	Case	Control
	n (%)	n (%)	n (%)	n (%)
Very active	2 (0.6)	0 (0.0)	1 (0.3)	0 (0.0)
Moderately active	30 (9.4)	4 (2.5)	12 (3.7)	2 (1.2)
Not active	58 (18.1)	25 (15.6)	20 (6.3)	7 (4.4)
Total	90 (28.1)	29 (18.1)	33 (10.3)	9 (5.6)

IGT = impaired glucose tolerance, IFG = impaired fasting glycemia

Table 3: Distribution of serum lipids in case and control subjects by gender

Lipid type (mmol/L)	Cases			Controls		
	Males	Females	P value	Males	Females	P value
Total cholesterol	4.5 (0.96)	4.7 (0.83)	0.05	4.5 (0.91)	4.7 (0.99)	0.187
HDL cholesterol	1.45 (0.43)	1.45 (0.45)	0.57	1.58 (0.40)	1.5 (0.39)	0.197
LDL cholesterol	2.5 (0.99)	2.7 (0.87)	0.62	2.4 (0.97)	2.7 (0.97)	0.902
Triglyceride	1.23 (0.52)	1.26 (0.55)	0.055	1.2 (0.51)	1.2 (0.51)	0.092

LDL = low-density lipoprotein, HDL = high-density lipoprotein

Data are means $\pm$ standard deviation (SD)

Table 4: Comparison of the clinical and metabolic characteristics of the case and control populations

Variable	Cases, <i>n</i> = 320	Controls, <i>n</i> = 160	<i>P</i> value
Age			
Male	35.7 (11.2)	36.7 (9.8)	0.250
Female	40.7 (12.7)	40.9 (10.4)	0.450
BMI			
Male	24.0 (4.6)	25.5 (4.8)	0.012*
Female	28.9 (6.3)	28.2 (5.4)	0.191
WC			
Male	84.5 (12.8)	88.5 (12.9)	0.014*
Female	91.3 (12.7)	90.1 (11.5)	0.232
SBP			
Male	120.3 (13.6)	122.2 (14.4)	0.167
Female	121.8 (15.4)	123.9 (19.3)	0.123
DBP			
Male	77.7 (9.5)	78.7 (9.9)	0.140
Female	79.3 (15.4)	81.9 (12.8)	0.090
FPG			
Male	4.6 (1.1)	4.4 (1.1)	0.100
Female	5.0 (1.5)	4.8 (1.7)	0.177
2HrPP			
Male	6.3 (2.3)	5.9 (2.0)	0.100
Female	7.4 (2.9)	6.8 (2.7)	0.056
Total Cholesterol			
Male	4.5 (0.9)	4.5 (0.9)	0.500
Female	4.7 (0.8)	4.7 (0.9)	0.500
LDL cholesterol			
Male	2.5 (0.9)	2.4 (0.9)	0.216
Female	2.7 (0.9)	2.7 (0.9)	0.500
HDL cholesterol			
Male	1.5 (0.4)	1.6 (0.4)	0.039*
Female	1.5 (0.5)	1.5 (0.4)	0.500
TGD			
Male	1.2 (0.5)	1.2 (0.5)	0.500
Female	1.3 (0.6)	1.2 (0.5)	0.093

WC = waist circumference, SBP = systolic blood pressure, DBP = diastolic blood pressure, BMI = body mass index, HDL = high-density lipoprotein, LDL = low-density lipoprotein, FPG = fasting plasma glucose, 2HrPP = 2 hour Post Prandial, TGD = Triglyceride

the increasing prevalence of diabetes mellitus worldwide,^[2] the number of FDRs of persons with type 2 diabetes, and thus an increased risk of developing diabetes, will also increase, which means that identifying risk factors associated with susceptibility to diabetes becomes increasingly important.

This study showed a relatively higher prevalence of IGT at 28.1% (34.9% in females and 20% in males, $P = 0.004$) among the cases compared with controls at 18.1% (21.4% in females and 14.5% in males, $P = 0.44$). This finding is similar to prevalence of 20% in Western European population of adults^[12] (25 years and older). In the US, the prevalence of IGT varies from 15.6% to 20.3%,^[13,14] which is similar to the findings among controls in this study.

The International Diabetes Federation (IDF) reported prevalence of 26% in United Arab Emirates,^[15] which is similar to the finding in this study.

The prevalence observed in this study is also similar to what was observed in Dar-es-Salaam, Tanzania (20%).^[16] The extrapolated prevalence in Nigeria according to US Census Bureau International data base was 21.3%,^[17] which was similar to the prevalence among controls. In Jos, North Central Nigeria, the reported prevalence was 26%^[18] among FDRs, which is similar to the findings in this study.

The finding of higher prevalence of IGT among females is similar to the report of IDF expert committee^[19] and the observation by Bakari and Onyemelukwe^[20] in Northern Nigeria.

The prevalence of IFG in this study was found to be 10.3% and 5.6% among study and control subjects, respectively. This is lower than the reported prevalence of 12.6%^[21] and 16.3%^[22] from Denmark and France, respectively. In the US, a prevalence of 12.1%^[13] was reported, which is higher than what was observed in this study. The subjects in the American study were older than those in the index study (40 vs. 25 years), which may explain the difference. Studies from Asia (India, 11%;^[23] China, 12.7%;^[24] and Iran 17.3%^[25]) also reported higher prevalence rates compared with this study. These studies from Asia were larger multicenter epidemiological studies and these may account for the higher prevalence rates observed.

On the African front, Amoah *et al.*^[26] reported a prevalence of 5.5% in indigenous Ghanaian subjects aged 25–75 years, which is similar to the findings among controls. In Jos, North Central Nigeria, a prevalence of 9.3%^[18] was reported among the cases, which is similar to the finding in this study.

This study found a higher prevalence of IGT among the non-active cases compared to the physically active ones (18.1% vs. 10%, $P = 0.005$). Similarly, among controls, the prevalence of IGT was lower in those that were physically active (2.5%) compared to those that were inactive (15.6%), $P = 0.0001$. It is, however, interesting that subjects and controls that engaged in exercise had significantly lower prevalence of IGT than those who did not engage in exercise.

The higher risk of IGT, IFG, and eventually diabetes associated with the cases was amplified in the presence of overweight, obesity, and sedentary lifestyle. Obesity was associated with T2DM in the general population as well.

This study also found higher serum lipid level in both study and control subjects with IGT, which concurs with previous report.^[18,25,27] This is most likely because IGT and diabetes are associated with a higher prevalence of additional cardiovascular risk factors, which collectively result in a

high-risk profile. Indeed, insulin resistance seems to play a major role in dyslipidemia in subjects with both normal and abnormal glucose tolerance,^[28] and appears to be the common element accounting for the cluster of atherogenic metabolic abnormalities found in the metabolic syndrome (IGT, hypertension, and dyslipidemia), which confers a high risk for cardiovascular disease.

The strength of this study borders on its design and contents. However, it is limited in its ability to allow definite generalizations to be made, and sequence of risk factor development could not be inferred. The study was designed to assess IGT (and by extension pancreatic beta cell function); insulin sensitivity and secretion are better indicators of pancreatic beta cell function and they can easily be assessed using homeostasis model assessment method, and this requires measurement of plasma insulin. Unfortunately, insulin assay could not be carried out as a part of this study.

Another limitation is that the study and control subjects were informed about the results and those diagnosed with T2DM are being followed up in the diabetes clinic but those with IGT and IFG were only counseled on appropriate lifestyle modifications as the study was not interventional.

CONCLUSION

In summary, the prevalence of IGT and IFG was higher in female cases (FDR of persons with T2DM) than that in controls of similar age and sex. Independent risk factors most closely associated with IGT were abnormal WC and WHR, FDR, elevated SBP, and DBP, whereas those closely associated with IFG were abnormal WC, FDR, and elevated SBP. These results emphasize the importance of controlling for all known diabetes risk factors, especially overweight and obesity, in FDR of people with T2DM.

These findings may prove useful in identifying a specific group of population at particular risk of developing metabolic disturbances known to predispose to cardiovascular disease and strongly support the regular screening of FDR of persons with T2DM.

Future studies on this topic may need to involve the assessment of the beta cell function so as to ascertain the level of secretory defect among the affected and perhaps genetic studies to establish the link between the patient and the FDR for better understanding and prospect of intervention.

Ethical approval

Informed consent was obtained from all individual participants of this study.

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Conflicts of interest

There are no conflicts of interest.

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